AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Chemistry Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 6, 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

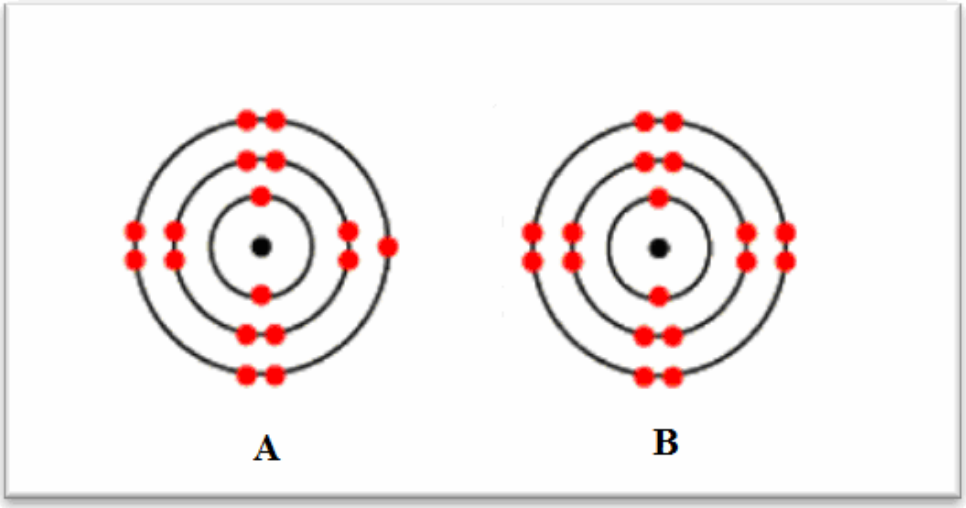
   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

The figure below shows the atomic structure of an element.



i. Identify which figure represents the chloride ion and also give the definition of an ion. (2 Marks)

ii. What charge will be present on a chloride ion? Explain giving a reason. (2 Marks)

Q.2. (Total 5 Marks)

a. A group of the periodic table is given below.

2 He 4.0026
10 Ne 20.179
18 Ar 39.948
36 Kr 83.80
54 Xe 131.30

Identify the group of the periodic table. Also explain why a group has similar properties. (2 Marks)

b. Which properties tend to decrease from top to bottom within a group? Give your answer with the help of an appropriate reason. (3 Marks)

Q.3. (Total 6 Marks)

a. Identify the following as ionic, covalent and co-ordinate covalent compounds. (4 Marks)

S.No	Compound	Type
1.	Ammonia	
2.	Cesium chloride	
3.	Hydrogen chloride	
4.	Sulphur dioxide	

b. What happens to the rate of evaporation when a liquid is heated? Also give a reason to explain your answer. (2 Marks)

_____

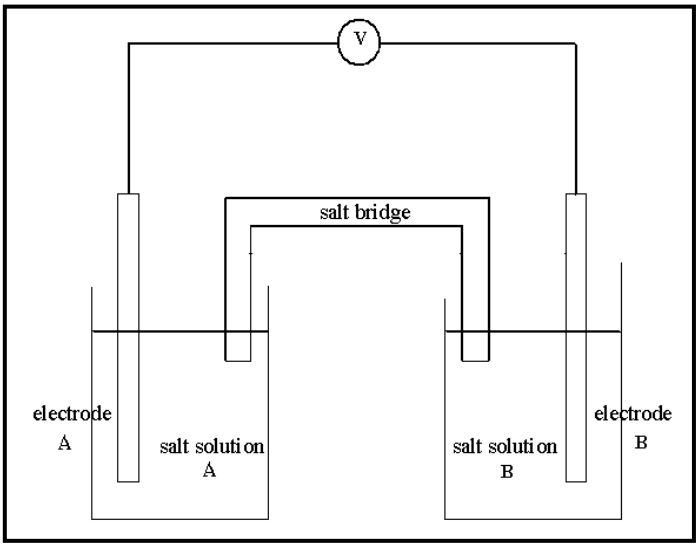
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Q.4. (Total 6 Marks)

a. An electrochemical cell is shown in the diagram. Students try using zinc, lead, magnesium and copper as electrodes.



i. Which TWO metals do you expect to give the highest reading on a voltmeter placed in the circuit? Explain your choice with the help of an appropriate reason. (2 Marks)

_____

_____

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ii. If copper and zinc are used as electrodes, what would you expect to see happening round the copper electrode? (1 Mark)

_____

_____

b. Differentiate between the following types of solution. Also give ONE example. (3 Marks)

S.No	True Solution	Colloidal Solution
1.		
Example		

Q.5. (Total 4 Marks)

a. Classify the following elements as alkali and alkaline earth metals in the grid. (2 Marks)

Elements:	**Al**	**Ca**	**P**	**Na**

Alkali Metal	**Alkaline Earth Metals**

b. What happens when the above alkali and alkaline earth metals react with halogens? Answer the question with the help of a balanced chemical equation. (2 Marks)

S.No	**Metals**	**Reaction With Halogens**
1.	**Alkali Metal**	**With Fluorine**
2.	**Alkaline Earth Metal**	**With Chlorine**

Q.6. (Total 5 Marks)

EITHER

a. Balance the following chemical equations with the help of inspections and trial and error method.

1. $Al_{(s)} + HCl_{(aq)} \rightarrow AlCl_{3(aq)} + H_{2(g)}$
2. $Zn_{(s)} + Fe_2O_{3(s)} \rightarrow Fe_{(s)} + ZnO_{(s)}$
3. $C_2H_5OH_{(l)} + O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$
4. $NO_{2(g)} + O_{2(g)} + H_2O_{(l)} \rightarrow HNO_{3(l)}$
5. $Na_2CO_{3(s)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

OR

b. Describe briefly the structure of an atom with the help of its sub atomic particles.

Q.7. (Total 5 Marks)

EITHER

a. Describe the formation of co-ordinate covalent bond by using an example of ammonium chloride.

OR

b. Write down any FIVE differences between the crystalline and amorphous solids.

Q.8. (Total 5 Marks)

EITHER

a. Write any FIVE differences between electrolytic and voltaic cell.

OR

b. How many millilitres of aqueous 2M KCl solution must be diluted with water to prepare 100 ml of aqueous 0.4M KCl? What do you mean by diluting a solution?
(Hint:Atomic mass of K= 39 amu)

END OF PAPER

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